

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: a4eb29ffe2d11f65

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : dsd.2021-12-02T16_44_54_046.wav
File SHA-256 Digest : a4eb29ffe2d11f65e04128ef5203ea2cdcab930c93dd30410fef83d00a443cb6
Payload File Size : 4,782,956 bytes (4670.9 KB)
Analysis Timestamp : 2026-09-19T08:38:43Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	74999.0 Hz	Baseband Nyquist Rate
Center Frequency (F_c)	+36.500 kHz	Spectral Centroid Offset
Estimated Baud Rate	2.67 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	54.9 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	72.93 kHz	Cumulative PSD 99% Energy
Estimated SNR	12.51 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	5.35 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	-9107.09 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0203	Wiener Entropy Index (0.0 – 1.0)

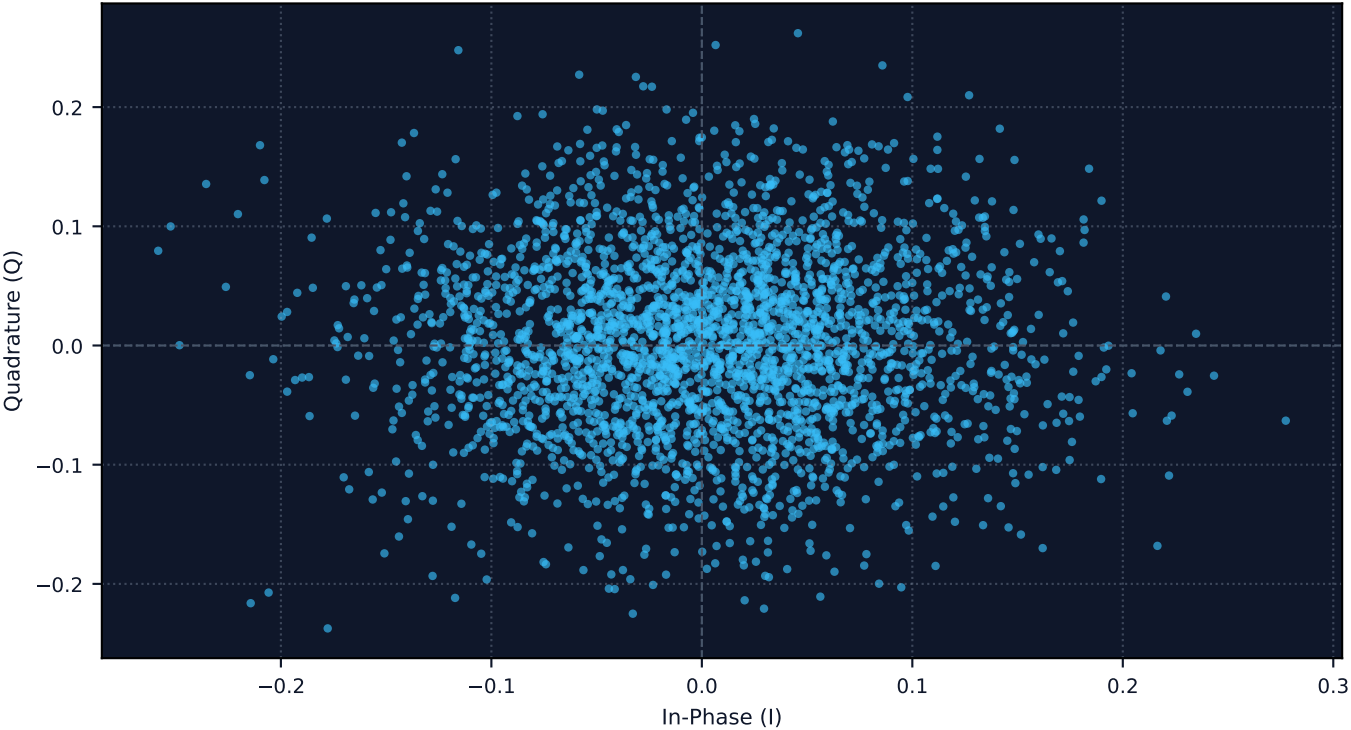
PREDICTED MODULATION : AM-DSB

CLASSIFIER CONFIDENCE : 84.67% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: BPSK | Recovered Bits: 149,460 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [AM-DSB]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	754	0x0000	0	18581 B	CRC MISMATCH	PARTIAL_RECOVER
#2	1133	0x0000	1	18534 B	CRC MISMATCH	PARTIAL_RECOVER
#3	1293	0x0000	2	18514 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1312	0x0000	3	18511 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1343	0x0000	4	18507 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 1E8A858BF0EA601F07CAE5F1B9BA169D [FRAME 1 PAYLOAD]